NETWORK VISIBILITY · AI CLASSIFICATION · QOS

Understand the network. Spot suspicious flows.

A network monitoring and decision-support system for connected devices

Presented by Your Name · Department · Date

Project Aim

What are we trying to achieve?

Give network administrators clearer visibility into device traffic, identify suspicious flows using machine learning, and support network management with QoS recommendations and optional firewall controls.

> In simple terms: monitor network activity, identify traffic that may be suspicious, and help an administrator decide what to do.

What to say: “The main aim is to make it easier to understand activity on a shared network. NetInsight collects information from connected devices, analyzes traffic, and presents useful results in one place.”

Why this project?

The challenge on a shared network

Many devices

Several users and devices share the same hotspot or local network.

Limited visibility

It can be difficult to see which device is generating unusual traffic.

Slow response

Without timely information, investigating and responding can take longer.

Administrators need a single view of device status, network activity, and possible threats.

What to say: “On a shared network, it is not always obvious which device is responsible for unusual traffic. This project brings monitoring and threat information together so it is easier to investigate.”

Project Objectives

Five practical goals

1. Collect device and network-flow telemetry through edge agents.
2. Classify traffic into threat categories using an XGBoost model.
3. Display device, traffic, and alert information in a dashboard.
4. Calculate bandwidth allocation recommendations using constrained optimization.
5. Support optional, controlled Windows Firewall restrictions for qualifying flows.

What to say: “These objectives describe the path from data collection to analysis and presentation. Firewall enforcement is optional and must be explicitly enabled.”

How the System Fits Together

A simple three-part view

1 · Edge Agent

Collects host statistics and packet-header information from a monitored device.

›

2 · NetInsight Server

Receives telemetry, builds flow records, runs classification and optimization logic.

›

3 · Dashboard

Shows device status, network metrics, classified flows, alerts, and recommendations.

Optional Windows Firewall enforcement is applied by the edge agent only when explicitly configured.

What to say: “The agent gathers information and sends it to the server. The server processes it, and the dashboard makes the results visible to an administrator.”

Main Components

Tools used in the project

Edge agent
Python, Scapy, and host telemetry

Backend
Django REST APIs

Database
SQLite for project records

Threat classifier
XGBoost machine-learning model

QoS optimizer
CVXOPT linear programming

Web interface
Dashboard for network and alert information

Optional enforcement
Windows Firewall on supported agents

What to say: “These components work together: the agent supplies network information, the Django server processes it, and the dashboard presents the results.”

How Traffic Is Analyzed

From packet headers to a visible result

01 · CAPTURE

The agent captures packet-header details and host telemetry.

02 · GROUP

The server groups packet records into network flows.

03 · CLASSIFY

The classifier evaluates flow features and predicts a category.

04 · RECORD

The system stores the label, confidence, and flow details.

05 · DISPLAY

The dashboard shows flow and alert information.

> A classification is an estimate from the model; it should be reviewed as evidence, not treated as proof of an attack.

What to say: “The project preserves the classification with the flow details, which makes it possible to inspect why a flow appeared in the dashboard.”

Threat Classification

XGBoost trained on CICIoT2023 prepared splits

Input features

- Average packet size (AVG)
- Rate
- Header length
- Packet number feature
- Protocol type

Output categories

- Normal
- DoS / DDoS
- Mirai Botnet
- Reconnaissance
- Brute Force

What to say: “The model evaluates prepared flow features and returns one of five categories. The categories include normal traffic and four groups of suspicious activity.”

Model Evaluation

Saved results on the prepared dataset

99.43%	98.87%	98.70%
Training accuracy	Validation accuracy	Test accuracy

Dataset split: 50,000 samples in each of training, validation, and test sets.

> These are saved evaluation results on prepared data. They do not guarantee the same performance on every real network or live traffic stream.

What to say: “The saved test-split accuracy is 98.70%. This describes evaluation on the prepared dataset and should not be interpreted as a guarantee for all live network traffic.”

QoS and Bandwidth Optimization

Recommendations within network capacity limits

What the optimizer does

- Uses available capacity and configured limits.
- Solves a constrained bandwidth allocation problem.
- Checks solution optimality using KKT conditions.
- Produces allocation recommendations.

Important boundary

Making capacity available does not guarantee equal bandwidth shares for all devices.

Guaranteed allocation requires appropriate router or network equipment controls.

What to say: “The optimizer calculates recommendations within capacity limits and verifies the solution. Actual sharing still depends on the network hardware and its configuration.”

Controlled Firewall Enforcement

Optional, scoped, and disabled by default

Before enforcement

- Explicit server and agent opt-in is required.
- Flow must have qualifying XGBoost evidence.
- Repeated packet evidence is required.
- Supported enforcement runs on Windows agents.

For the demonstration

- Use an existing registered agent.
- Send a small, paced replay to a private receiver you control.
- Restrict the rule to that specific demo flow.
- Report success only after agent acknowledgement.

> This controls a matching outbound flow; it does not block an entire device or guarantee bandwidth distribution.

What to say: “Firewall enforcement is deliberately opt-in. The demonstration is scoped to a private receiver, and the project reports a block only when the agent confirms its rule.”

Dashboard and Demonstration

What the reviewer can see

Devices

Connected agents and their reported status.

Network

Throughput, capacity, and performance metrics.

Security

Classified flows, alerts, and reported enforcement state.

Show: agent telemetry → classification → dashboard result → optional rule acknowledgement.

> Have dashboard screenshots ready as a backup if the live network or firewall setup is unavailable.

What to say: “I’ll show how telemetry becomes a classified flow and appears in the dashboard. I’ll claim enforcement only if the agent explicitly acknowledges the matching rule.”

Testing and Verification

Checks run on the project

12 / 12	0	0
Django tests passed	Django system-check issues	Missing migrations reported

Regression coverage includes flow thresholds, XGBoost filtering, demo-event validation, dashboard metrics, and firewall-rule matching.

> Automated tests verify application logic; they do not guarantee that every laptop’s live capture and firewall setup will work identically.

What to say: “The automated Django suite passed all 12 tests, the system check found no issues, and the migration check found no missing migration.”

Limitations and Future Work

Current limitations

- Firewall enforcement supports Windows agents.
- Live capture depends on the device and capture setup.
- Dataset evaluation may not represent every real network.
- Equal sharing requires suitable router or network controls.

Future improvements

- Test with more devices and network environments.
- Evaluate additional datasets and live traffic.
- Extend enforcement support to other platforms.
- Integrate with router-level QoS controls.

What to say: “The current version has clear platform and deployment limits. Future work can expand testing, improve real-world evaluation, and connect with more network equipment.”

IN SUMMARY

One system for network visibility and informed response.

Edge monitoring · XGBoost classification · QoS recommendations · Optional firewall controls

Thank you

Questions?